

HR Analytics SQL Project Report

Objective

The primary objective of this project is to conduct HR analytics using Structured Query Language (SQL) to extract insights from a Human Resources dataset (HRDataset_v14). The goal is to evaluate workforce structure, salary trends, recruitment efficiency, performance distribution, attrition risks, and other key metrics to support data-driven decision-making in HR.

Dataset Overview

The dataset includes detailed information about employees such as:

- Employee details: Employee_Name, Department, Position
- Compensation: Salary
- Demographics: Age
- Employment timeline: Hire_Date, TermDate, EmploymentStatus
- Performance: PerformanceScore, LastPerformanceReviewDate
- Engagement: EngagementSurvey
- Hierarchy: ManagerName
- Recruitment source: RecruitmentSource

Key Analyses and SQL Insights

1. Workforce Composition

- Listed all employees with their departments and positions.
- Counted the number of employees in each department.
- Showed how many employees were hired through each recruitment source.

2. Employment Status & Termination Analysis

- Identified all currently active employees.
- Counted the number of active vs. terminated employees.
- Retrieved employees who left the organization within 1 year of joining.
- Listed top 3 departments with the highest number of terminations.
- Flagged employees with no termination date who are not marked as active.

3. Salary and Compensation

- Calculated the minimum, maximum, and average salary across the company.
- Listed the top 5 highest-paid employees.
- Computed average salary per department and per position.
- Identified departments where average salary exceeds the company average.
- Compared each employee's salary with the average salary in their department using window functions.

4. Performance & Retention

- Counted employees under each performance score.
- Listed employees who received an "Exceeds" performance score.
- Identified employees with strong performance and low salaries (potential retention risk).
- Flagged high-risk employees with good performance but low engagement and low compensation.
- Calculated the percentage of each performance rating by hire year.

5. Recruitment Source Effectiveness

- Analyzed average salary per recruitment source.
- Determined which recruitment source delivers higher-paid talent.
- Calculated turnover rate per source to identify the most stable recruitment channel.

6. Engagement Metrics

- Calculated average engagement survey scores by department.
- Cross-referenced departments with low engagement and high attrition.

7. Tenure and Experience

- Calculated tenure (in years) for each employee.
- Grouped employees into tenure buckets: <1 year, 1-3 years, 3-5 years, 5+ years.
- Evaluated year-wise hiring trends.
- Analyzed salary growth trends based on hire year.

8. Managerial Reporting and Promotions

- Counted how many employees report to each manager.
- Identified employees promoted more than once (assumed promotion tracking column).

9. Time Series Analysis

- Displayed month-over-month trends of hires and terminations.
- Highlighted employees who have not had a performance review in over 12 months.

Tools & Techniques Used

- Database Language: MySQL
- SQL Concepts: Aggregations, Filtering, Grouping, Subqueries, Window Functions
- Functions Used: COUNT, AVG, MIN, MAX, CASE, RANK(), TIMESTAMPDIFF, DATE_FORMAT, IFNULL, CURDATE
- Date Handling: Tenure calculation, performance review gap analysis, hiring/termination trends.

Conclusion

This SQL-based HR analytics project reveals critical insights into employee distribution, compensation fairness, performance trends, and potential retention issues. These queries provide a strong foundation for making data-informed HR decisions. Integrating these results into BI dashboards (e.g., Power BI or Tableau) would further enhance interpretability and visual storytelling.